

Chap 5: Crime Hacking

- o What is “Hacking”?
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- o Hackers as Security Researchers
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What is “Hacking”?

Definition

- Hacking refers to gaining **unauthorized access to computer systems, networks, or data.**
- It involves exploiting **software vulnerabilities, weak passwords, or security gaps.**

Types of Hackers

- **White Hat Hackers** – Ethical hackers who improve security.
- **Black Hat Hackers** – Malicious hackers who steal or damage data.
- **Gray Hat Hackers** – Hackers who access systems without permission but may not intend harm.

What is “Hacking”?

Common Techniques

- o Phishing attacks
- o Malware installation
- o Password cracking
- o Network intrusion

Impact

- o Data theft
- o Financial losses
- o Privacy violations
- o National security risks

Hacktivism (Political Hacking)

Definition

- o Hacktivism is the use of hacking techniques to **promote political or social causes**.

Objectives

- o Protest against governments or corporations
- o Promote political ideologies
- o Expose corruption or injustice

Common Activities

- o Website defacement
- o Data leaks
- o Distributed Denial-of-Service (DDoS) attacks

Examples

- o Political protest hacking groups
- o Campaigns against censorship or surveillance

Debate

- o Supporters: Digital activism
- o Critics: Cybercrime and illegal disruption

Hackers as Security Researchers

Role

- o Many hackers work as **ethical security researchers**.

Purpose

- o Identify vulnerabilities in systems
- o Prevent cyberattacks
- o Strengthen cybersecurity defenses

Activities

- o Penetration testing
- o Vulnerability scanning
- o Responsible disclosure of security flaws

Benefits

- o Improves software security
- o Protects users from cyber threats
- o Helps organizations strengthen defenses

Hacking as Foreign Policy

Cyber Warfare

- o Governments sometimes use hacking for **national security or intelligence purposes**.

Uses

- o Espionage (stealing classified information)
- o Disrupting infrastructure
- o Influencing elections

Examples

- o Cyber attacks on power grids
- o Government-sponsored cyber espionage

Concerns

- o International conflict
- o Cyber arms race
- o Difficulty in attribution

The Law: Catching and Punishing Hackers

Cybercrime Laws

Governments have introduced laws to punish cybercriminals.

Key Legal Measures

- o Computer misuse laws
- o Cybercrime investigation units
- o Digital evidence collection

Challenges

- o Hackers hide identities
- o Cross-border crimes
- o Rapidly evolving technology

Punishments

- o Fines
- o Imprisonment
- o Asset confiscation

Identity Theft and Credit Card Fraud

Identity Theft

- When someone **steals personal information** to impersonate another person.

Common Targets

- Bank accounts
- Credit cards
- Social security numbers

Methods

- Phishing emails
- Data breaches
- Malware

Consequences

- Financial loss
- Damaged credit history
- Privacy violations

Stealing Identities

Definition

- o Criminals obtain personal data such as:
 - o Names
 - o CNIC numbers
 - o Bank details
 - o Passwords

Sources

- o Social media
- o Data breaches
- o Public databases

Uses

- o Fraudulent transactions
- o Opening fake accounts
- o Identity impersonation

Responses to Identity Theft

Personal Protection

- o Strong passwords
- o Two-factor authentication
- o Regular credit monitoring

Institutional Measures

- o Fraud detection systems
- o Secure payment technologies
- o Customer verification processes

Government Role

- o Cybercrime laws
- o Consumer protection regulations
- o Cybercrime investigation agencies

Biometrics

Definition

- Biometrics uses **unique biological characteristics** for identification.

Examples

- Fingerprints
- Facial recognition
- Iris scans
- Voice recognition

Advantages

- Strong authentication
- Hard to duplicate
- Convenient for users

Concerns

- Privacy issues
- Data misuse
- Biometric database breaches

Whose Laws Rule the Web?

Global Nature of the Internet

- o Internet operates **across national borders**.

Problem

Different countries have **different laws** regarding:

- o Privacy
- o Cybercrime
- o Content regulation

Example

A website hosted in one country may violate laws in another country.

Key Question

Which country's law should apply?

When Digital Actions Cross Borders

Cross-Border Cybercrime

- o Many cybercrimes occur **across multiple countries**.

Challenges

- o Jurisdiction issues
- o International cooperation
- o Evidence sharing

Solutions

- o International treaties
- o Cybercrime cooperation agreements
- o Global law enforcement collaboration

Libel, Speech, and Commercial Law

Libel

- Publishing false statements that harm someone's reputation.

Online Speech

- Social media raises issues of:
 - Freedom of speech
 - Hate speech
 - Fake news

Commercial Law Online

- E-commerce regulations
- Consumer protection
- Digital contracts

Culture, Law, and Ethics

Different Cultural Values

- o What is acceptable online varies by culture.

Examples

- o Free speech laws differ worldwide
- o Privacy expectations vary

Ethical Issues

- o Responsible technology use
- o Respect for privacy
- o Avoiding harm through digital actions

Potential Solutions

Improving Cybersecurity

- o Strong encryption
- o Advanced security systems

International Cooperation

- o Global cybercrime laws
- o Information sharing between countries

Public Awareness

- o Cybersecurity education
- o Safe internet practices

Technological Solutions

- o AI-based threat detection
- o Multi-factor authentication
- o Secure digital identity systems

Case Study: The Cyber Attack on GlobalTech Corporation

Background

- o GlobalTech Corporation is a multinational technology company that provides cloud services, e-commerce platforms, and digital payment systems. Millions of customers worldwide use its services to store personal data, make online purchases, and conduct financial transactions.
- o One day, GlobalTech discovered that its systems had been compromised by a group of hackers. The attackers gained unauthorized access to customer databases and stole thousands of users' personal identities, including credit card details, addresses, and login credentials.

Case Study: The Cyber Attack on GlobalTech Corporation

- Shortly after the breach, some of the stolen data appeared online. Investigators later found that the attackers were part of a hacktivist group protesting against GlobalTech's alleged involvement in government surveillance programs.
- The hackers claimed that their actions were meant to expose unethical practices and raise awareness about privacy violations. However, the breach also resulted in serious consequences for customers, including identity theft and credit card fraud.
- Authorities from several countries began investigating the case because the hackers operated from different countries, the company's servers were located in multiple regions, and the victims were spread across the globe.

Case Study: The Cyber Attack on GlobalTech Corporation

1. What type of hacking occurred in this case?

o Answer:

This case involves **unauthorized access to computer systems and databases**, which is a form of hacking. The hackers exploited vulnerabilities in GlobalTech's network security to gain access to sensitive customer data.

2. How does this case demonstrate hacktivism?

o Answer:

The attackers claimed they were protesting against GlobalTech's alleged cooperation with government surveillance programs. Using hacking to promote a political or social cause is known as **hacktivism**.

Potential Solutions

3. How could ethical hackers or security researchers have helped prevent this attack?

o **Answer:**

Ethical hackers could have:

- o Conducted **penetration testing**
- o Identified vulnerabilities in the company's systems
- o Reported weaknesses through **responsible disclosure**
- o This would allow GlobalTech to fix security flaws before attackers exploited them.

4. How can hacking be used as a foreign policy tool?

o **Answer:**

Governments sometimes support cyber operations to gather intelligence, influence political situations, or disrupt foreign systems. If a government had secretly supported the hackers in this case, it could represent **cyber warfare or cyber espionage**.

Potential Solutions

5. What laws apply to catching and punishing the hackers?

Answer:

Cybercrime laws such as:

- o Computer misuse laws
- o Data protection laws
- o Fraud and identity theft laws
- o Law enforcement agencies can investigate the attack, collect digital evidence, and prosecute those responsible.

6. What is identity theft and how did it occur in this case?

Answer:

Identity theft occurs when criminals steal personal information and use it to impersonate someone else. In this case, hackers stole customers' **names, credit card numbers, and addresses**, which could be used for fraudulent transactions.

Potential Solutions

7. What is credit card fraud and how might criminals use the stolen data?

Answer:

Credit card fraud occurs when someone uses another person's credit card details without permission. Criminals could:

- o Make unauthorized purchases
- o Transfer money
- o Create fake accounts for financial transactions

8. What steps should victims take in response to identity theft?

Answer:

Victims should:

- o Report the fraud to their bank immediately
- o Change passwords
- o Monitor credit reports
- o Enable two-factor authentication
- o Report the incident to cybercrime authorities

Case Study: The Cyber Attack on GlobalTech Corporation

9. How could biometric security help prevent such incidents?

Answer:

- Biometric systems such as **fingerprints, facial recognition, or iris scans** can strengthen security because they rely on unique physical characteristics. Even if passwords are stolen, biometric verification can prevent unauthorized access.

Case Study: The Cyber Attack on GlobalTech Corporation

10. Why is it difficult to determine whose laws apply in this case?

o **Answer:**

The internet operates globally. In this case:

- o Hackers were in different countries
- o The company's servers were located internationally
- o Victims were worldwide
- o This creates **jurisdiction problems**, making it difficult to determine which country's laws apply.

11. How do digital actions crossing borders complicate cybercrime investigations?

o **Answer:**

Cross-border cybercrime investigations are difficult because:

- o Countries have different legal systems
- o Evidence may be stored internationally
- o Cooperation between law enforcement agencies is required

Case Study: The Cyber Attack on GlobalTech Corporation

o 12. How could libel, speech, and commercial law issues arise in this situation?

o Answer:

If the hackers published allegations about GlobalTech online, it could raise **libel issues** if the claims were false.

Additionally, commercial laws may apply because customer financial information and e-commerce transactions were affected.

13. How do culture, law, and ethics influence perspectives on this case?

o Answer:

Different societies may view hacktivism differently. Some may see it as **digital protest**, while others consider it **illegal cybercrime**. Ethical perspectives also vary regarding privacy, activism, and corporate responsibility.

Case Study: The Cyber Attack on GlobalTech Corporation

14. What are potential solutions to prevent such cyber incidents in the future?

o Answer:

Possible solutions include:

- o Strong cybersecurity systems**
- o Encryption and multi-factor authentication**
- o Ethical hacking and security audits**
- o International cooperation against cybercrime**
- o Public awareness about cybersecurity risks**
- o**